

Continued Fractions

In 1597, the great Johannes Kepler wrote about what he considered geometry's "two great treasures": the Pythagorean theorem and a number that has since come to be called the *golden ratio*. Often denoted by the Greek letter *phi*, the golden ratio is equal to about 1.618, and Kepler was only one of dozens of great thinkers who have been entranced by it. Like pi and a few other famous constants, such as the exponential base *e*, phi has a tendency to show up in unexpected places. People have found phi in many places in nature, and have painstakingly documented where it occurs in fine art, as in the annotated version of the Rokeby Venus shown in [Figure 5-1](#).

In [Figure 5-1](#), a phi enthusiast has added overlays that indicate that the ratios of some of these lengths, like b/a and d/c , seem to be equal to phi. Many great paintings have a composition that's amenable to this kind of phi-hunting.